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Global why

Overview

- How has globalization become a salient theme?
- While manifestations and expressions of globalization vary widely, underlying trends are world population growth, growing population density and growing connectivity.
- Trade and migration, conquest and pilgrimage produced geographical and cultural knowledge. Collecting global data is as old as the first charts of sea and land routes and world maps.
- What is the demand and what is the supply of global knowledge?
- Globalization involves long term transformations as well as recent accelerations.
- Globalization gifts of art, technology and knowledge have long sprawled across regions and more recently, fashion, style, film and media worlds.
- Demands of globalization move up the ladder of attention, from foreign affairs to global problems.

How has globalization become such a salient theme? While manifestations of globalization vary widely, underlying trends that have long been in motion are world population growth, growing population density and growing connectivity. Connectivity has been growing since time immemorial. Trade and migration, conquest and pilgrimage required and generated geographical and cultural knowledge. Collecting global data is as old as the first charts of sea and land routes and world maps. This chapter gives an overview of global turns, considers premises of global thinking and establishes links between connectivity and globalization.

Global gifts, global demands

Global cultural interplay and mixing are as old as the hills, are ongoing and unstoppable. Fashion has long adopted styles and fabrics from across continents and regions, well beyond the ethnic chic and tribal turn of past decades. Multicentric couture goes back as far as the Roman Empire (assimilating Greek and Phoenician, Persian and Egyptian styles), the Middle Ages and colonial times. European football is huge in Asia and rising fast in China. The widening range of products available on supermarket shelves and the variety of culinary styles sprawls and follows people's travel habits. Medicine, healing and spiritual practices have long crossed civilizational boundaries. Apothecaries in medieval Europe carried a Moor's head as signboard to profile the oriental origins of medicinal herbs.

In art, biennales have proliferated across regions; new art markets, museums and galleries are mushrooming; fluctuating art prices reflect regional trends. Architects from across the world build iconic structures of a postnational character. Western architects design new mosques in Arab countries. New art venues give rise to new criteria of art (chapter 7).

Newsreel was multi-sited early on. Movies have increasingly become multi-sited, filmed in different countries with local actors, language, cultures (such as the Jason Bourne series and The International). Global movies have become a regular fixture. Multi-country filming started in spy and James Bond movies, spread to action movies, then moved across film genres and has become ordinary, also in TV production and Netflix (such as *Sense8* and *Money Heist*).

Media worlds have long sprawled across regions, not just BBC world radio, also film and TV with the popularity of Bollywood and Hong Kong martial arts movies in Africa, Turkish soaps in Saudi Arabia, Brazilian telenovelas in Latin America and Lusophone Africa and the Korean wave in East Asia. The K wave, Hallyu, in pop, soap operas, movies, TV shows, design and cosmetics is unstoppable. Bollywood and Hollywood have strengthened their cooperation. Hollywood now needs China for blockbuster box office numbers. Dalian Wanda has bought into Hollywood studios. The downstream production of books, film and media is increasingly decentered too.

While Amazon, Disney and HBO stream mainly American movie content, Netflix is a 'global platform'. Netflix's 'goal is to appeal to an increasingly global audience with an increasingly global slate'. Its shows are seen in more than 190 countries around the world. Netflix has production hubs in London, Madrid and Toronto and content teams on the ground in many other countries such as Mexico. According to the head of Netflix's non-English language original programming: 'when we go into a country, it really is about the storytelling—a creator with a vision from that country' (Lev-Ram 2018: 82). Netflix's programs have stirred local debates, in Argentina (a docu-drama about the death of attorney general Nisman), Brazil and India. A global platform that produces local stories with global production techniques and a global appeal is a novel kind of format and a harbinger of new level of global consciousness.

Global production and value networks have proliferated since the 1980s as have global supply chains for retailers and supermarket chains (Carrefour, Tesco, Walmart). Digital tech facilitates long-distance communication, containerization lowers transport cost and jet air travel continues to expand. The salience of the global and the growth of global mixing is not just trendy; it is structurally embedded in the world economy.

Major structural changes have been at work at the same time, in particular population growth, urbanization (Table 1.1) and technological change. Together these add up to a confluence that often goes under the heading of globalization, though much more is taking place. Globalization, then, is a stand-in for wider changes, an umbrella under which many changes are provisionally parked. These changes appear to be in overdrive especially since the mid-twentieth century. This is how people often view globalization.

Table 1.1 World population growth, share of urban population and population density over time

Year	World population (millions)	Urban population (% of total)	World population density (inhabitants/km2)[+]
1000	295[*]		1.98
1500	461[*]		3
1800	989[*]	2[±]	6.6
1900	1654[*]	15[•]	11.1
1970	3,700[+]	36.5[†]	24.8
1980	4,458[+]	39.2[†]	29.9

2000	6,145[+]	46.5[†]	41.2
2017	7,550[+]	54.2[†]	50.6

Sources[1]

Among major changes over recent decades, James Rosenau counts urbanization, greater mobility (travel, tourism, migration), education and environmental change. According to Rosenau, these changes give rise to a 'capacities revolution'. On account of increased information processing due to microelectronics, education, media, the growing complexity of large urban communities and growing mobility, a skills revolution is taking place. Dealing with greater urban and environmental complexity enhances intellectual functioning and induces 'a greater capacity to focus emotions' and a freeing up of the imagination. In his view, 'the skill revolution is world-wide in the sense that even as the analytically, emotionally, and imaginatively rich get richer, so do those who are poor in these respects get richer'. Part of this is an 'organization explosion'—'the global stage has become ever more dense with actors' (Rosenau 1999: 61, 1011-14).

Globalization, then, is a matter of long term transformations as well as recent accelerations. Considering we talk about globalization on two wavelengths, short term and long term, we should probably call this a skills evolution. Population growth, urbanization and technological change have accompanied human evolution throughout history, though not in a linear fashion. Bronze Age technological evolution (the plough and animal traction) enabled a surge in agricultural productivity, which spurred population growth, trade and an urban revolution (Goody 2006). Thus we can fine tune the assessment as follows: a) major structural changes go back at least 5000 years, b) they have been marked by periods of acceleration, slowdown and breakdown, c) they have been accompanied all along by skills evolution, d) the past two centuries have been periods of major acceleration as well as the period since the 1970s, the time that Rosenau and many others talk about.

It is a truism that the twenty-first century is more saliently global than the preceding century, which is noticeable in virtually any sphere—in economics, finance, advertising, media as well as in culture, politics and social movements. The global isn't new in any of these domains and in some areas goes back for thousands of years, but its salience is new. Global awareness is more widespread than before, including awareness of global problems (climate change, epidemics such as HIV Aids, bird flu, coronavirus, crime syndicates, terrorism, marine piracy, multinational corporations and economic crises). Migration and emancipation movements of women, minorities and migrant workers shape new publics and make new demands on the public sphere under headings such as multiculturalism and diversity. While these concerns constitute the demand side of global knowledge, global studies is on the supply side.

According to Mark Zuckerberg's Facebook manifesto, Building Global Community, 'Our greatest challenges also need global responses—like ending terrorism, fighting climate change, and preventing pandemics. Progress now requires humanity coming together not just as cities or nations, but also as a global community' (Zuckerberg 2017). These are examples of the demands that the acceleration and growing density of global interdependence pose. Inequality, poverty, migration, surveillance, technologies such as 5G and drones, epidemics and Covid-19 are others.

In the old days, foreign affairs and diplomacy was an affair of gentlemen, often of aristocratic background. In medieval Europe, aristocracies intermarried across borders, dynastic marriages sealed the fate of kingdoms, aristocratic orders such as the Golden Fleece and the Order of the Garter glued aristocratic bonds and alliances. The Church clergy was 'transnational' and so were humanist scholars such as Erasmus and Renaissance artists such as Dürer. Trade alliances such as the Hanseatic League established commercial bonds.

The nineteenth-century rise of nation states gave rise to 'foreign affairs'. In the course of the twentieth century international affairs followed foreign affairs and was in turn succeeded by 'global affairs' and 'global issues'. Contemporary terminology has moved on to 'global problems'. The demands of the global keep moving up the ladder of attention. As we face global problems and challenges we do so mostly with national institutions. International institutions are underfunded and are viewed as rivals by hegemonic powers. The chronic disparagement of the United Nations in American media and government is an example. Many problems are global, but media and public spheres are mostly national or provincial in outlook. We often face global problems with national or provincial attitudes.

Global data

Social sciences and humanities fulfill service functions. They would not exist unless they meet social demand and provide interested parties with relevant information, perspectives and skills. This includes analytical functions and critique—which discuss which types of data are relevant, which categories, concepts and classifications matter, examine structures and institutions, and question epistemological premises and cultural assumptions. Global studies, likewise, meets a social demand. Adopting a sociology of knowledge approach, a sociology of global knowledge can start out by identifying the social demand that global knowledge seeks to meet.

The rise of global studies reflects the growing presence of the global. Simply count how often the adjective 'global' appears in news headlines and reports. Global studies reflect the growing pace, scope and intensity of global relations and effects. Global studies have been growing because of the exponential growth of global dynamics; it is a response to ramifying, intensifying and deepening processes of globalization. Global studies has been spreading because global relations and problems require a global approach, a need that is felt by governments, social forces, international organizations and corporations the world over.

In relation to this social demand, global knowledge represents the supply. We can distinguish three levels of global knowledge. The first level is global data and information per se; the wide array of diverse and sprawling data collected by all actors and institutions that have an interest in global information of some kind, such as international institutions, governments, corporations, social movements, media and foundations. Many data are collected to meet specific demand and are systematically organized according to limited purposes. Familiar sources of global data are the UN agencies, the World Bank, IMF, the CIA Fact Book, the Economist Intelligence Unit, regional development banks, university research centers, professional associations, etc. Other organizations and corporations seek global information of more specific types, such as the Peace research center in Oslo on conflicts and Oxfam International on inequality. In addition,

media and literature supply data that are diverse and may be impressionistic and anecdotal. This basic level of global data is constantly growing, wide in scope and fragmented.

The second level consists of globalization research, which has mostly been organized according to social science and humanities disciplines' legacies and theories. The third level is global studies as the integration of these bodies of knowledge. In sum, the three levels of global knowledge are global data—wide, yet fragmented; globalization research—influenced by discipline demarcations and theories; and global studies—the integration of the above and an approach in its own right.

Table 1.3 gives a sketch of global data—indicative, not exhaustive. The first column lists the agency of social demand and the relevant type of data; the second column indicates the knowledge supply that seeks to meet this demand. Each of these could be detailed in many subsets but this only seeks to give an impression of the sprawl and diversity of global data. What is listed is a sample (being exhaustive would serve little purpose).

Table 1.2 Global data

Social demand	Global data supply
Governments Internationalization, geoeconomics, geopolitics, crossborder movements, cultural flows, refugees	Comparative politics; regulatory and legal regimes; security, conflict flash points; crime, terrorism, human trafficking; demographics, studies of religion; surveillance systems
International institutions Environment; resource use; international development; demography, migration, urbanization; international law, ICC, Interpol; disease, public health	Sustainability and climate studies; development data, statistics such as literacy rates, Human Development Index; migration studies; nodal cities, infrastructure, cartels, money laundering
Organizations Professional associations, think tanks, educational institutions	Specialist knowledge networks, international conferences, e.g. physicians, epidemiologists, lawyers, security analysts, accountants, travel; higher education policies
Trade unions, ILO	Labor standards and regulations, labor rights regimes. Minimum wage, social protection floor
Media, ICT Markets, investments, cooperation, events	ICT hardware and use, satellites, fiber optic cables, regulatory regimes; media production networks, centers, flows
Sports International sports, Olympic Games, World cups, tournaments, sponsors, clubs	Global markets in soccer, golf, tennis, cricket; global talent searches
Art Auction houses, museums, dealers, collectors, artists, academies	Art markets, fairs, biennales, galleries, regional and transnational aesthetics, price trends and fluctuations
NGOs, social movements, foundations	

International NGOs. World Council of Churches, ecumenical organizations, World Social Forum	Aid policies and social demand; transnational activism, global reform from below, pressure points of social change; interfaith dialogue
Corporations Commodities, logistics, shipping, insurance. Global market share, joint ventures, global brands, advertising; offshoring, outsourcing, institutional arbitrage; technology, innovation, intellectual property, laws, piracy. Banks, hedge funds, institutional investors, credit rating agencies. World Economic Forum, Davos. Tourism, events management. Luxury goods markets, fashion, cosmetics	Transportation networks; financial fluctuations, currencies; price movements, regulatory regimes, business cultures; interlocking boards; patent regimes and enforcement; stock exchanges, credit ratings. Global culture, 'cool hunters', style; beauty pageants, contests. Luxury studies, global fashion.

Global narratives

Stories such as *A 1001 Nights* traverse worlds of space and imagination. Agatha Christie's classic thriller, *Murder on the Orient Express* and the movie versions owe much of their appeal to their traversing vast ancient lands. They are journeys that take us back to our subliminal memory and ancestral knowledge.

Among the 2,500 books in Richard Wagner's library in Wahnfried, his house in Bayreuth, Bavaria many are plays, comedies, literary classics, myths and legends. Wagner revived the idea of the Gesamtkunstwerk (total work of art) and transformed opera from light entertainment to powerful cosmic battles of good and evil, against a backdrop of ancient myths and legends (Young 2014). Wagner's operas such as Der *Ring des Nibelungen* (1874) are preludes to Tolkien's *Lord of the Ring*, to *Star Wars*, the Harry Potter series, *Game of Thrones* and science fiction novels and movies that unfold at the edge of the abyss, the precipice of apocalypse. Thus, legendary tales of Manichean struggles between good and evil continue to shape fantasy and media worlds.

According to an advertisement page in *Fortune* magazine, 'With more data than ever before, what we need is intelligence'. 'Reveal the hidden patterns that will change your business' promises the ad of a research firm that uses a streaming infrastructure.

The issue in contemporary globalization is not data and knowledge. Data are plenty, facts abound and big data join world encyclopedias of information. Google, Wikipedia and encyclopedias are a click away. Data on global trade, investment, poverty, migration, disease and so forth are reported on a daily basis and in reports of international institutions and agencies such as the IMF, Oxfam International and WHO. The weak link in contemporary globalization is narratives that organize data and index information in meaningful ways. Without stories that integrate them in coherent wholes, facts are merely *faits divers* (a classic newspaper heading for assorted news tidbits). In his play *Six actors in search of an author* (1921), Luigi Pirandello observes that facts by themselves are just empty sacks; they only stand up if they are filled with meaning.

Why narratives and not analyses? Because stories speak to emotion and imagination. Narratives focus the mind. Stories form the psychosocial tissue that drives and enables social cooperation. All social cooperation needs stories that focus attention, motivate and organize social interaction. Stories function at all levels, family, local, national, regional and global.

As Ben Anderson observed, nations are 'imagined communities' that have been cobbled together by means of transport and print media, education and the invention of traditions (1983). Nations are narrations (Bhabha 1990). Likewise, globalization is an assemblage of stories. At times we experience globalization directly, as in travel, fast fashion and fast food on high streets, but we often experience globalization through representations of various kinds, media, movies, reports and metrics. As was the case with nations and other imagined communities, globalization comes to us in the form of crisscrossing stories. Our global coexistence is a salad bar of stories.

Take finance. Because finance relies on credit (belief) which is a function of reputation, it is fundamentally a matter of representations, such as analysts' reports, market data, newspaper reports, corporate gossip as well as the equations of quantitative investors. Securities and stock values, futures and derivatives largely unfold through stories and in the imagination (Cameron and Palan 2004).

Stories experience four seasons. They are sown, germinate, rise, peak, unravel and decline. There are always stories coming and going and crisscrossing on the way. Most conversations in the public sphere are the jostling of stories. Newspaper headlines launch storylines and pinprick storylines. Stories work cognitively, emotionally and imaginatively. For stories to motivate and carry social cooperation they must be repeated often (as in advertising), generate familiarity (a shared story) and be backed up by delivery (some kind of material record). Delivery and its assessment hinge on representations. Stories come in many guises, myths, fables, scriptures, fairy tales, folklore, rituals, ceremonies, traditions, ideologies, paintings, architecture, national anthems and symbols. 'Every picture tells a story'. No social gathering occurs without a storyline (a birthday, wedding, celebration, funeral), which also applies on wider scales. Marx and Engels' Communist Manifesto, Gandhi's salt March, the Muslim hajj, Martin Luther King's 'I have a dream', Nkrumah's pan Africanism and at an international level, the Cold War and domino theory are examples of collective narratives.

Stories also represent collective learning, from the invention of fire and Homer's tales onward. Growing connectivity means a growing demand for narratives that organize and give meaning to connectivity. 'All the world's a stage' has been true all along. Now more than before, it is a global stage.

What we now call globalization went under earlier headings such as 'internationalization' and 'interdependence'. The 1970s was an era of international reform proposals such as the New International Economic Order initiated by the Dutch economist Jan Tinbergen (1976) and the New International Information Order endorsed by UNESCO. The pro-market turn of the 1980s sidelined these perspectives and gave rise to new global narratives. Keynesianism had replaced neoclassical economics, and efficient market theory replaced Keynesianism. The 'free market and democracy' and Washington consensus narratives gave cohesion to IMF (conditionalities) and World Bank programs (structural adjustment). The fall of the Berlin Wall signaled the triumph of liberal democracy. For consumers this was to usher in an era of a 'borderless world' and 'the world as a global duty free store' (Ohmae 1992). These narratives inspired trade pacts such as the World Trade Organization and regional cooperation such as NAFTA and ASEAN. The Club of Rome brought ecological concerns and Limits of Growth to the foreground. Global narratives inspire

institutions and organizations of cooperation. In the 2000s the language of globalization became digital and metamorphosed as interconnectedness and connectivity.

A sample of collective stories is Table 1.4. The stories are broadly familiar. To each there are background stories and subplots as well as specific organizational initiatives and changes over time. It goes too far to discuss this in detail. The point is to illustrate the collective importance of stories in focusing and organizing attention, also on the global stage. National, regional and global cooperation all refer to 'imagined communities'. The overview omits local and national narratives, of which there are many.

Table 1.4 Collective narratives—sample

Scale	Institutions	Narratives
Global	United Nations	Bretton Woods system, 1944 UN Charter, Declaration of Human Rights, 1948 New International Economic Order, 1974 UN Millennium Goals; Sustainable Development Goals, 2015 Paris Climate Accord, 2016
	International law	Nuremberg trial, International Court of Justice, International Criminal Court, War Crimes Tribunals
	GATT	General Agreement on Tariffs and Trade, 1948-1995
	WTO	World Trade Organization, 1995
	Non-Aligned Movement	Bandung 1955, South-South cooperation
	Club of Rome	Limits to Growth, 1972
	NIEO	New International Economic Order, 1974
	G20	Global systemic crisis, 2008. Seoul consensus, 2011
	BRICS	South-South cooperation, 2001. New Development Bank
	Investment banks	Emerging market investment memes
	World Economic Forum	'As America retreats, China advances', Davos, 2017
	China, BRI	Belt and Road Initiative, 2013
Hegemonic	United States (IMF, World Bank GATT, WTO)	Cold war, domino theory, containment, rollback NATO, Marshall Plan, Alliance for Progress Free market and democracy. Efficient market theory Washington consensus, 1990s 'The end of history', liberal democracy triumphs, 1989 Clash of civilizations, 1993 America First, 2016
Regional	Regional integration	European Union, ASEAN, Gulf Cooperation Council, Mercosur, etc.
	Developmental states	Asian tigers, the East Asian miracle, 1993. Pacific century
	Asian values	Neo-Confucian ethic
	21C socialism	Cuba, Venezuela, Kerala
From below	Social movements, NGOs	Abolitionism, women's movements, working class struggles, national struggles—19C Decolonization, Third World solidarity, 1968, peace movements, Via Campesina, battle of Seattle—20C

		Occupy Wall Street, Indignados, environmental justice, Extinction Rebellion, animal rights, Black Lives Matter—21C

While several of these stories claim global sway, they are mostly *regional stories* that seek to organize world order from the viewpoint of a zone or center of influence. Abundant knowledge of connectivity is available, but to access its meaning often means to subscribe to one or other hegemonic narrative. Through the nineteenth century, the British Empire and Europe were major global story tellers—the Enlightenment, stages theories, progress, the civilizing mission, national sovereignty, modernity, etc. In the twentieth century, the 'American Century', the United States was a major global storyteller, ranging from Washington to Hollywood and popular culture. Familiar story lines are modernization, Fordism, Coca-Cola, McDonald's, Walt Disney, Barbie and CNN.

There were counter narratives to the Cold War (such as the Bandung conference of 1955, the Movement of Nonaligned Countries and the Tricontinental Movement, based in Havana). Counter narratives to globalization also emerged. Globalization was cast as neocolonialism, Americanization, Coca-colonization, McDonaldization, west-toxification (in Iranian views) and neoliberal globalization. Asian values, the neo-Confucian ethic, Islamic values and Latin American cosmovisions (such as the Zapatistas' Alliance of Hope and *buen vivir*) served as counterpoints.

In twenty-first century the tables have turned. Emerging societies and the global South *welcome* globalization and free trade while in the US and part of Europe globalization is often viewed with discomfort. Emerging societies provide new narratives such as the Seoul consensus (2011) and China's grand Belt and Road Initiative (2013), a new Silk Road. International institutions advance Millennium Development Goals and Sustainable Development Goals and international cooperation advances accords such as the Paris Climate Accord. In the US, economic nationalism and 'America First', anti-immigration and demographic chauvinism are counter narratives to 'globalism'. Trade tariffs, securitization measures and border walls also emerge in parts of Europe (chapters 10, 11).

Globalization, then, is a salad bar of stories that are different in different parts of the world, with different meanings and different time cycles. Each form of connectivity such as trade, finance, migration, military reach and big tech comes with winners and losers (the distribution of which changes over time) and is contentious. The assessment of gains and losses is itself a function of representations. Global narratives are not just descriptions but also prescriptions and arenas of contention. The social construction of reality also takes place on a world scale.

According to the Polish philosopher Henrik Skolimowski, for the 'participatory mind' evolution is taking a reflexive turn: 'we are evolution conscious of itself' (1994: 92). Our growing engagement with global trends and concerns, with global inequality, with not just national and regional but also global development, with climate change and species' futures signals 'evolution conscious of itself'. The notion of the Anthropocene marks a recognition of sapiens' impact on other species and the planet.

Cornerstones of Hegel's *Philosophy of History* are consciousness (Geist, spirit) and agency (the ability to act upon consciousness). Consciousness, awareness of what is going on in the world and agency are keynotes of what Hegel called the awakening of the spirit of history (Hegel 1840,

2001). In this perspective, the difference between being an object of history and a subject of history is consciousness and agency. Observed Marx in the *Eighteenth Brumaire of Louis Bonaparte*, 'Men make their own history, but they do not make it as they please… but under circumstances existing already'. Claude Lévi-Strauss paraphrased: 'Men make their own history but they do not know they are doing it' (cited in Loyer 2018).

Are stories, also global stories, just-so stories or do they also represent collective learning? In a book on development theory I redefine development, not as improvement of living standards, enlargement of choices or enhancement of capabilities, but as a process of *collective learning*, socially distributed and shared trial and error processes (Nederveen Pieterse 2010: 191). Globalization, too, can be viewed as a process of collective learning and global narratives and global studies articulate this learning. One of the social demands that globalization makes is awareness of achievements and challenges of global coexistence, its historical depth and evolutionary momentum. Facing our global rendezvous consciously, part of reflexivity is recognizing the role of narratives of globalization, intellectually and emotionally. There is a narrative character and a collective imagination to our collective coexistence. Global awareness, of course, is also a story about stories. It features humanity as a subject of history, 'evolution conscious of itself'. A challenge of global consciousness is to examine global stories that meaningfully organize our collective experience. The chapters that follow explore this challenge.

Overview

- Many problems are global but media and public spheres are mostly national in outlook.
- Movements of women, minorities and migrant workers shape new publics and make new demands on the public sphere under headings such as multiculturalism and diversity.
- Global awareness is growing, including awareness of global problems.
- Levels of global knowledge are global data—fragmented; globalization research—demarcated by discipline outlooks; and global studies—the integration of the above.
- Global studies is a response to ramifying and deepening processes of globalization.
- The weak link in globalization is narratives that organize information in meaningful ways.
- We mostly experience globalization through media representations, reports and metrics.
- Globalization comes to us in the form of crisscrossing stories.
- Global narratives are often regional stories that portray the world from the viewpoint of a center of influence.
- Each form of connectivity comes with winners and losers and is contentious.
- Globalization is an assemblage of stories in different parts of the world with different meanings and time cycles. Global narratives are not just descriptions but also prescriptions and arenas of contention. Stories also represent collective learning.

Notes

[1] [+] Calculations based on land area of the world, 148.94 million square km. [*] Goldewijk, K. K., A. Beusen and P. Janssen 2010 Long-term dynamic modeling of global population and built-up area in a spatially explicit way: HYDE 3.1, *The Holocene* 20:565–73. [+] UN Department of Economic and Social Affairs 2017 *World Population Prospects: the 2017 Revision* Vol 1. Available at https://esa.un.org/unpd/wpp/Publications/Files/WPP2017_Volume-I_Comprehensive-Tables.pdf. [±] UN 2001 *Istanbul+5: Habitat Agenda*. New York, June 6-8. Available at http://www.un.org/ga/Istanbul+5/booklet4.pdf. [•] Spence et al., eds. 2008. [†] Figure is for 2016, World Bank.